

Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	1	of	8

NSP/004/102 (OHI 2) Guidance on erecting Single or 'H' poles

1. Purpose

The purpose of this document is to describe the approved techniques for the handling and erection of wood poles for use on the CE Electric UK Distribution System.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
RTN / 001 / 500 / 741 / 004	18.02.97	GN 74-04 Guidance on the equipment and techniques for the manual handling of overhead line wooden poles.
RTN / 001 / 500 / 741 / 005	09.12.96	GN 74-05 Guidance and general principles for erecting Single or 'H' poles using both manual or mechanical methods.
RTN / 001 / 500 / 741 / 006	28.07.97	GN 74-06 Guidance Note for General Principles of Pole Alignment
OHI 2	Oct 2000	Pole Erection

2. Scope

This document has been prepared to provide guidance on pole handling and erection techniques. It provides high level guidance on the allowable techniques available for pole erection. More detailed guidance on the specific techniques can be found in the specific work instruction or task instruction identified in the table.

3. Technical Specification

3.1 Background

Wooden poles are heavy and difficult to handle manually. Good techniques and safe lifting practices will not only make the job much easier but will reduce the risk of injury. When moving poles you must follow the guidance given in this note to avoid injury to yourself and others.

Before starting work:

- Carry out a risk assessment to make sure the task and work site is safe.
- Establish a safe working zone using signs, cones and barriers, fencing or ropes as appropriate.
- Check the pole for signs of damage in accordance with OHI 12 (NSP/004/112)

Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	2	of	8

Precautions

- Always wear your PPE, as a minimum this should include overalls with the sleeves rolled down, safety footwear, safety helmet and industrial gloves. If in doubt about any aspects of your work, contact your supervisor.
- Wood poles are treated with one of two preservatives, “creosote” or “CCA” both of which can cause skin irritations. Additionally splinters and trapped skin are common injuries which will be avoided if users wear gloves.

Notes

- Where practical use mechanical means such as a “Strimech” in preference to manual handling of poles.
- Where manual handling is unavoidable, use mechanical aids to help reduce the risk of injury e.g. the use of the pole bogie (wheel trolley) and pulling and lifting equipment such as “Turfors”, ropes and/or bars when moving poles,

3.2 Tools & Equipment

Make a visual inspection of all lifting equipment before use, check to make sure the colour code marking is valid. Choose lifting equipment SWL to suit the total weight being lifted, typical weights (for guidance only) of ENA TS 43-40 construction steelwork and wood poles are shown in the table below.

3.3 Typical Equipment and Pole Weights

3.3.1 Steelwork weights

Type of Support	Weight kg
Single crossarm intermediate	60
Double crossarm intermediate	110
Double crossarm section / terminal	120
Narrow “H” pole section	120
Wide “H” pole section	140
33kV Terminal	200

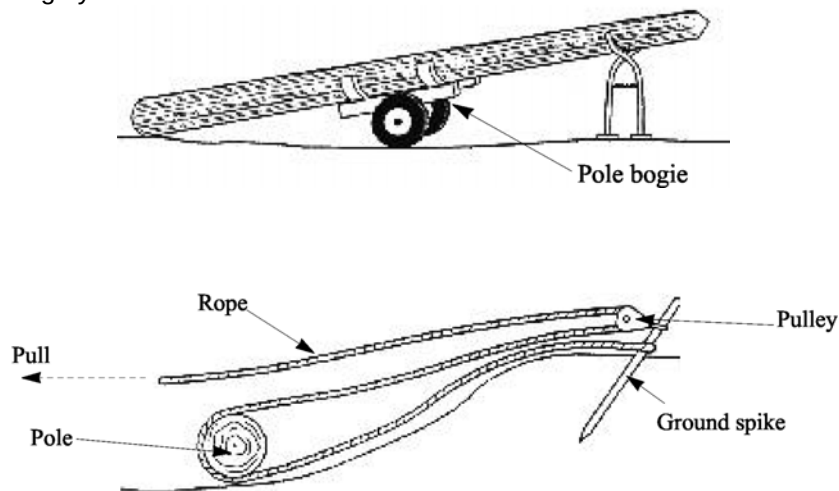
3.3.2 Typical Pole weights

Length Metres	Length Ft/ins	Light Kg	Medium kg	Stout kg	Extra Stout kg
8.0	26-3	143			
8.5	27-11	157	213	305	
9.0	29-6	163	229	335	
9.5	31-2	183	249	371	
10.0	32-10	198	259	395	427
10.5	34-5	213	284	427	498
11.0	36-1	234	310	472	528
11.5	37-9	264	335	498	569

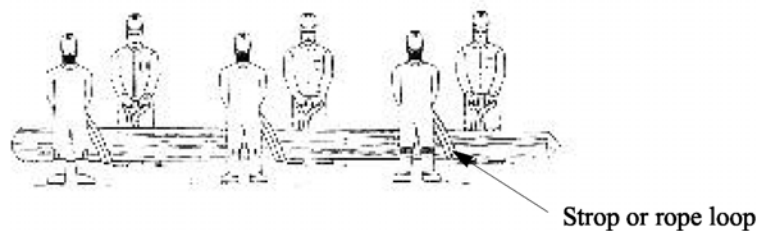
Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	3	of	8

Length Metres	Length Ft/ins	Light Kg	Medium kg	Stout kg	Extra Stout kg
12.0	39-4	284	356	539	620
13.0	42-8	315	417	599	671
14.0	45-11		467	686	762
15.0	49-3		554	762	859
16.0	52-6		630	848	
17.0	55-9		721	950	
18.0	59-1		828	1110	
20.0	65-7		996	1340	
22.0	72-2		1110	1550	
24.0	78-9			1630	

3.4 Lifting and moving by hand



- When lifting by hand use as many members of the team as possible and use rope loops or strops to keep your fingers and hands free from trapping.
- Always have only one person in control of the lift to call out lifting instructions.



Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	4	of	8

3.5 Pole erection techniques

Poles shall be erected by one of the following methods:

Pole erection method	Limitations	Existing YEDL Task/Work Instruction	Existing NEDL Task/Work Instruction
Mechanical			
Strimech pole handler (Single poles)	Suitable for erecting single poles with maximum combined pole/steelwork weights of 1000kg (16m stout)	WI 74-09	-
Strimech pole handler ('H' poles)	Suitable for erecting 'H' poles with maximum combined pole/steelwork weights of 1500kg (14m Stout)	WI 74-06	-
Mechanical Excavator (front bucket/bull horns)	Suitable for erecting single poles with maximum combined pole/steelwork weights of 1000kg (14m stout grade)	WI 74-05	04/034
Mechanical Excavator (back actor)	Suitable for erecting single poles with maximum combined pole/steelwork weights of 1000kg	-	04/033
Rotaclaw	Suitable for erecting single poles - SWL's related to combined pole/steelwork weights and lifting radius (ranging from 560 – 960kg)	WI 74-04	04/015
Crane Mounted Vehicle	Suitable for erecting single or 'H' poles - SWL's related to combined pole/steelwork weights and lifting radius (Dependant upon vehicle lifting capacity)	WI 74-01	04/016
Manual			
Falling Derrick	Suitable for erecting single poles with maximum combined pole/steelwork weights of 1000kg	WI 74-02	04/010
Pike Method	Suitable for erecting single poles sizes up to 12m Medium Grade	WI 74-03	04/011
Shearlegs (Swedish Lifter)	Suitable for erecting single poles with a maximum combined pole/steelwork weights of 350kg	-	04/012
Falling Derrick 'H' Pole	Suitable for erecting 'H' poles with maximum combined pole/steelwork weights of 1000kg (14m Medium grade)	WI 74-07	04/013
Falling Derrick 'H' Pole or Gin Pole	Suitable for erecting 'H' poles with maximum combined pole/steelwork weights of 3000kg (20m Stout grade)	WI 74-08	04/014
Via Existing Pole	Suitable for erecting single or 'H' poles with maximum combined pole/steelwork weights of 1000kg.	-	04/018

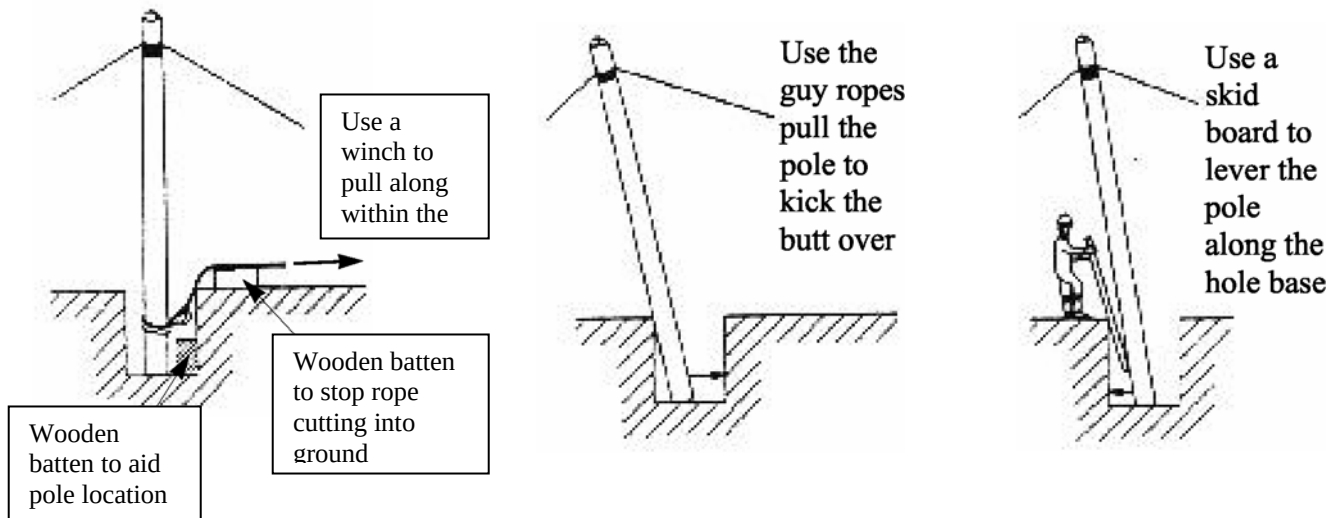
3.6 Pole alignment techniques

Following the planting of a pole into a pole hole, care shall be taken to ensure that the pole is vertical and in line with the overhead line route. After aligning of the pole, the pole shall be temporarily supported to prevent movement until the excavation is reinstated.

Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	5	of	8

3.6.1 Moving a pole in line

Always use a mechanical pole handler as the preferred method of adjustment following the appropriate work instruction. Where this is not practical use manual methods as detailed below.

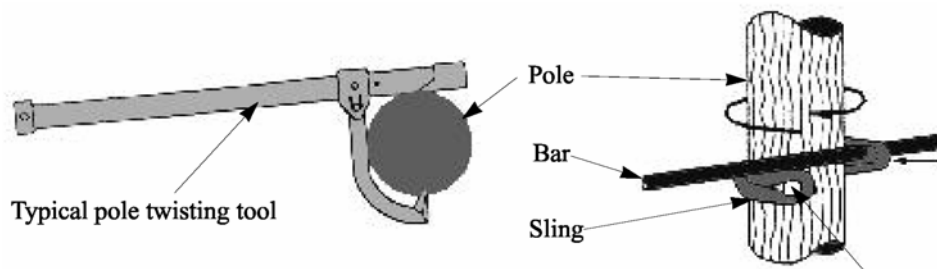


Plumbing a pole:-

- Always plumb a pole before the hole is backfilled.
- If after backfilling the pole it is found not to be plumb, then excavate the ground around the pole to a depth of at least 600mm, loosen the soil below this level around the pole base and attach guy ropes to secure the pole whilst plumbing.
- Take care not to stress the pole unduly

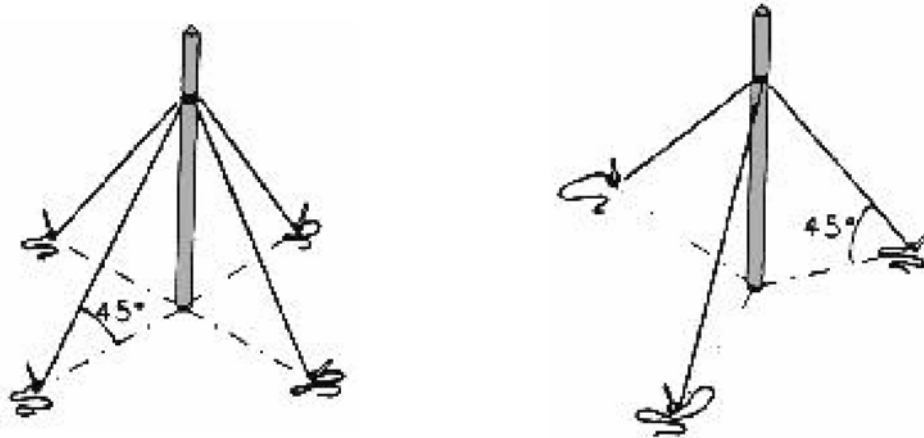
Twisting a pole:-

- If the pole is fitted with kicking blocks you must fully excavate these before attempting to twist the pole.
- If the pole is to be twisted before backfilling of the pole hole, then either the pole twisting tool method or the bar and sling method.



- If the pole is to be twisted after backfilling then only use the bar and sling method as this is less damaging to the pole.

Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	6	of	8



Securing a pole:-

- When securing a pole use four guy ropes evenly spaced (at 90° intervals) around the pole. If this is not possible or practicable then use three guy ropes at (at 120° intervals).
 - Wherever possible attach guy ropes to the pole before it is erected. If you need to attach guy ropes to the pole after erection use a hydraulic bucket lift as a means of access and attach the guy ropes at height or, use a 'rope support' attachment for live line tapping sticks and follow the procedure below.
1. Tie one end of the rope loosely around the base of the pole using a running bowline.
 2. Rest the rope in the guides of the support and attach the support to the tapping sticks.
 3. Raise the rope to the desired height and pull the free end of the rope, tightening the guy rope knot around the pole.
 4. Release the rope support and make off the guy to its anchor
 5. Repeat as necessary.

Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	7	of	8

4.0 References

4.1 External Documentation

Reference	Title
-----------	-------

4.2 Internal documentation

Reference	Title
WI 74-01	Work Instruction for erecting a single pole using a semi-mobile Crane
WI 74-02	Work instruction for erecting a single pole using a 1000 kg 'A' frame derrick and winch.
WI 74-03	Work instruction for erecting a single pole using 'pole pikes'
WI 74-04	Work instruction for erecting a single pole using a 'Rotaclaw' pole handler
WI 74-05	Work instruction for erecting a single pole using a 'bull horns' pole handler
WI 74-07	Work Instruction for erecting 'H' poles using a 1000kg 'A' frame derrick and winch.
WI 74-06	Work instruction for erecting 'H' poles using a strimech 'H' pole handler
WI 74-08	Work Instruction for erecting 'H' poles using a 3000kg 'A' frame derrick and winch.
WI 74-09	Work instruction for erecting a single pole using a strimech pole handler
04/010	Work specification – Erection of single pole using a falling derrick 'A' frame
04/011	Work specification – Pole erection – Pike method
04/012	Work specification – Pole erection – Pole raising shearlegs (Swedish lifters)
04/013	Work specification – Erection of H pole Falling Derrick 'A' Frame.
04/014	Work specification – Pole erection – Gin pole method
04/015	Work specification - Pole erection, using 'Rotaclaw' power rotating
04/016	Work specification - Wood pole replacement using a vehicle mounted crane
04/018	Work specification - Pole erection using existing pole
04/033	Work specification - Wood pole erection – mechanical excavator using back actor
04/034	Work specification - Wood pole erection – mechanical excavator using front bucket

5. Definitions

Term	Definition
------	------------

Document reference		NSP/004/102					
Version:-	1.0	Date of Issue:-	Jul 06	Page	8	of	8

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	31/05/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/06/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/06/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	21/07/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
D Sillito	Major Project Manager South	D Sillito	17/06/06
S Gardner	OHL Construction Manager	S Gardner	16/06/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
G Earl	Primary Engineering Projects Manager	G Earl	19/06/06